

How Stable is SHAP under Oversampling?

A Study of Explanations in Imbalanced Scenarios

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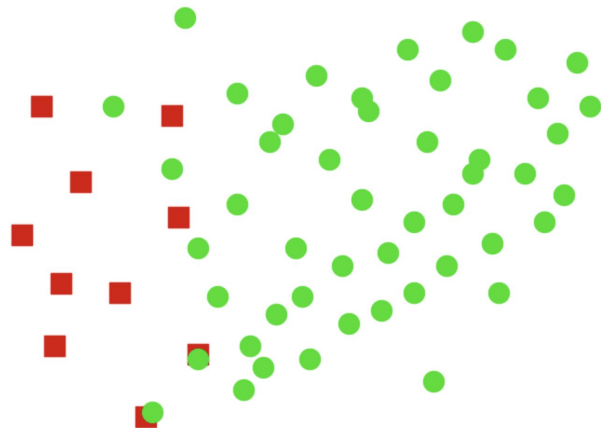
Introduction

Class imbalance is common in classification problems, and **oversampling techniques** are a widely adopted solution.

Data-level resampling methods such as **SMOTE**, **ADASYN** and their variants are now standard tools.

Oversampling techniques alter the learned decision logic of classifiers and, consequently, the explanations derived from them.

Rebalanced models exhibit shifted concepts and **altered variable importance** patterns.





Motivations

There are several publications investigating effect of data rebalancing on explainability demonstrating the interest in the topic

Studies about different practical problems lead to different (and even opposite) results about effects of oversampling over explanation stability

There is a lack of studies aimed to approach the problem in a general way

research domain

Finding correlations between data oversampling in imbalanced datasets and explanation stability in a context detached from any particular application



XAI Literature about oversampling

Patil, Aum, Aman Framewala, and Faruk Kazi. 'Explainability of SMOTE Based Oversampling for Imbalanced Dataset Problems'. 2020 3rd International Conference on Information and Computer Technologies (ICICT), March 2020, 41–45. <https://doi.org/10.1109/ICICT50521.2020.00015>.

“The results showcased [...] that the oversampling does not alter the features correlation of the original dataset”

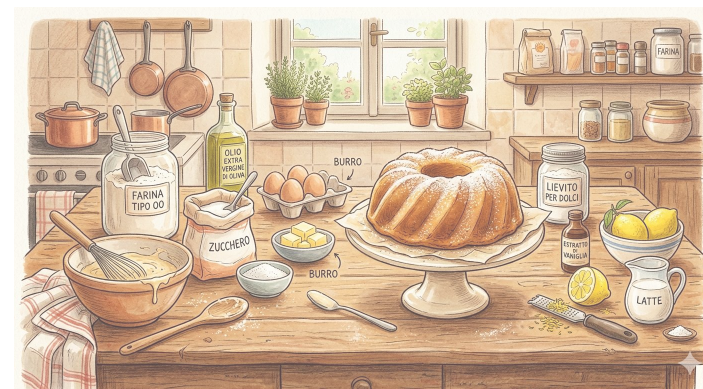
Rai, Teena, Jun He, Jaspreet Kaur, et al. 'Evaluating XAI Techniques under Class Imbalance Using CPRD Data'. Frontiers in Artificial Intelligence 8 (November 2025): 1682919. <https://doi.org/10.3389/frai.2025.1682919>.

“the consistency of LIME and SHAP explanations decreased as class imbalance increased when evaluated against a balanced dataset” [for a specific clinic dataset]

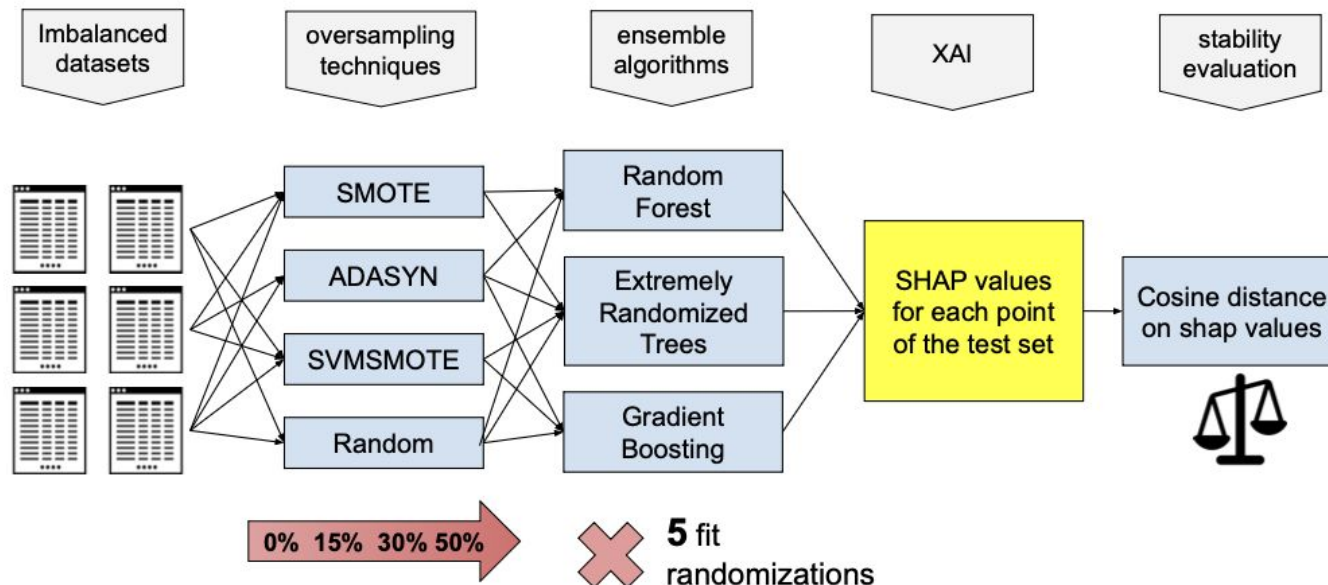
methodology - overview

ingredients

- tabular binary imbalance datasets (25 imbalanced binary classification datasets with heterogeneous properties, 336-20,000 instances, 6-617 features, 6:1-130:1 imbalance ratios)
- oversampling (ADASYN, ROS, SMOTE, and SVM SMOTE)
- 4 levels of rebalancing (0%, 15%, 30%, 50%)
- tree based classifiers (XTR, RF, and GBC)
- local SHAP
- SHAP stability (cosine distance)

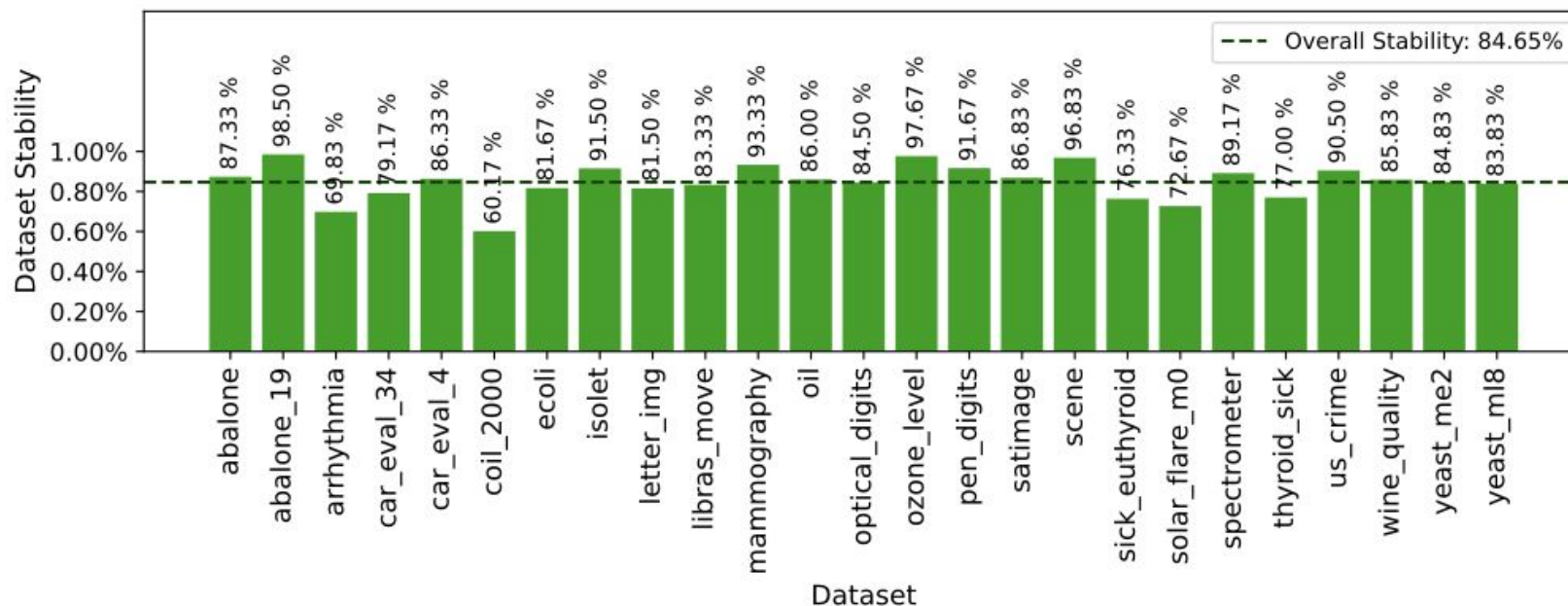


methodology - Experimental Design



Results - rq 1

How do local SHAP explanations change as class balancing increases by oversampling?



results - rq 2

Is it possible to identify a classifier and resampling combination that yields more stable explanations across different random initialisations?

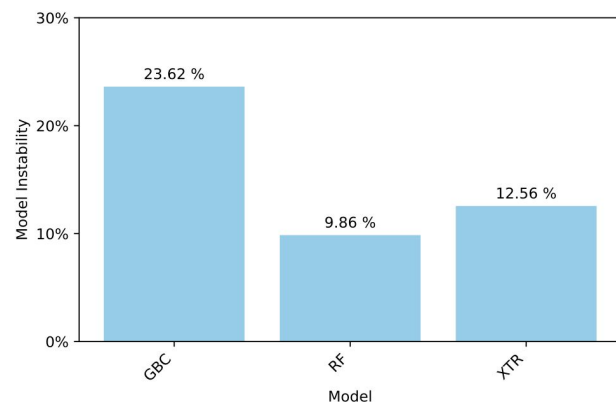
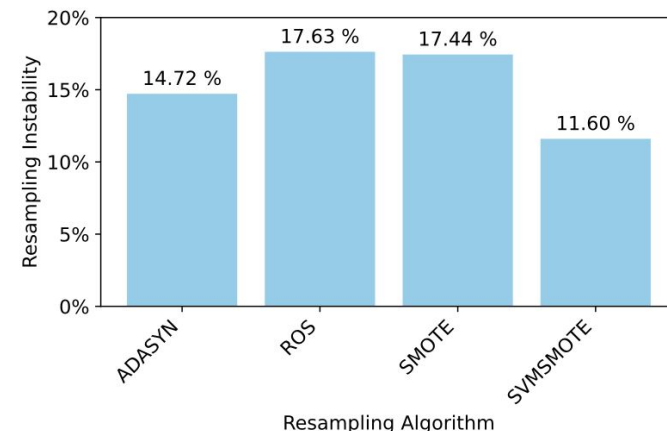


Figure 3: Model-level instability indicator (higher is worse).

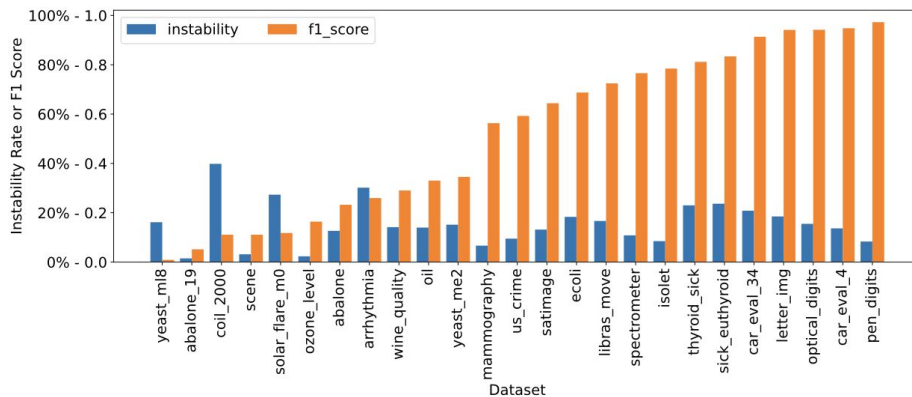
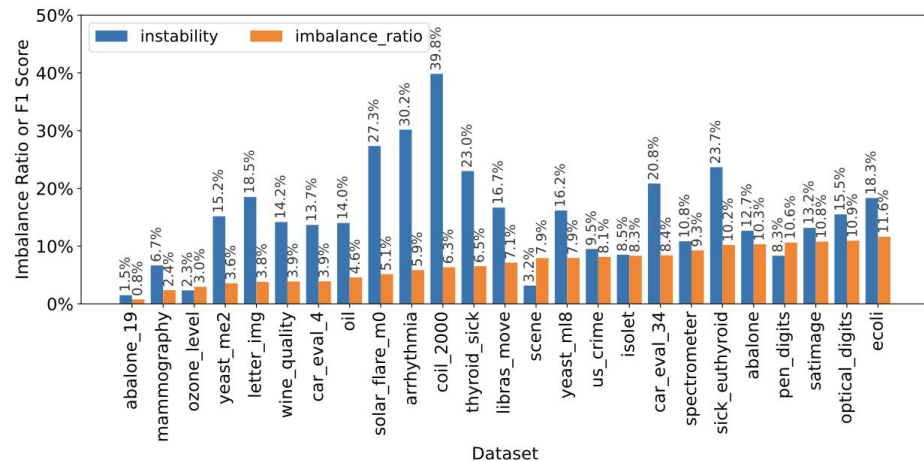


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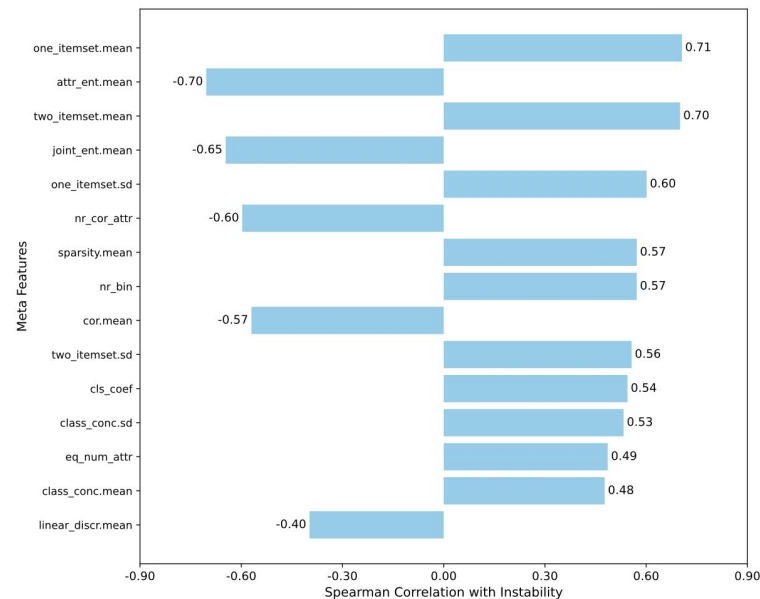
results - rq 3

Which intrinsic dataset characteristics are associated with SHAP instability under oversampling?



results - rq 3

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limitations

- The study involve only relative measurements of SHAP stability
- The experiments are restricted to tree-based classifiers
- meta-feature analysis does not establish causality

extensions

- expand upon these results by considering additional datasets, model families and imbalance countermeasures
- validate conclusions using alternative stability metrics
- develop meta-learning tools that predict instability risk from dataset descriptors to support explanation-aware pipeline selection.
- formulate theoretical hypotheses that explain the observed correlations as well as the deviation from the mean of some datasets or algorithms



Conclusion

The findings indicate that common oversampling strategies do not systematically destabilize SHAP explanations.

Some correlations between SHAP stability over oversampling and specific algorithms were observed; we also observed a correlation between SHAP stability and specific datasets meta features.

All the results are confirmed by a large number of experiments but need more investigations into local behaviors that are not aligned with the mean results.

Questions?

Thank you!

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more details about
the experiment



<https://github.com/sodomacouniv/how-stable-is-SHAP-under-oversampling>